

Proton Chair Eigenmodes in the MMU: From Internal Rotor Frequencies to Hydrogenic Space Response

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Abstract

In the Methane Metauniverse (MMU), the proton is modelled as a tetrahedral resonator whose three internal axes w_2, w_3, w_4 generate electric, torsional, and volumetric response [5, 9]. At the coarse-grained level this structure closes into a hexagonal “chair” ring, formed by six coupled segments associated with the sub-tetrahedra along w_2, w_3, w_4 , consistent with the microstructure identified in [12, 13].

In this note we construct a minimal six-mass, six-spring model of the proton chair, using only the MMU stiffness rules $k_2 \sim 2\alpha\hbar c/a^3$, $k_3 \sim \hbar c/a^3$ as derived in [5, 9, 16], together with the proton charge radius $a_p \simeq 0.84$ fm and the proton mass. Solving the eigenvalue problem yields a triplet of internal rotor frequencies with characteristic energies of order $E_1 \simeq 290$ MeV, $E_3 \simeq 500$ MeV, $E_5 \simeq 580$ MeV, without any additional fit parameters.

We interpret these modes as the intrinsic “motor” of the proton: a set of strong-scale torsional oscillations that drive the surrounding MMU lattice [4, 12]. In the continuum limit the chair dynamics reproduces the radial Laplacian with Coulomb term, leading to the hydrogenic Laguerre solutions previously derived in the MMU framework [8, 15]. In this picture the Schrödinger equation becomes the macroscopic space response to the proton rotor, while the bound electron acts as a negative-charge indicator that makes these standing waves visible through spectral lines [3, 15].

1 MMU Proton as Hexagonal Chair Rotor

In the MMU spacetime model the proton is realised as a positively oriented UR-cell (tetrahedron) with an internal hierarchy of three sub-tetrahedra along the deformation axes w_2 (electric), w_3 (torsional), and w_4 (volumetric) [5, 9]. The triangular faces of these sub-tetrahedra close into the minimal oscillatory structure with D_6 symmetry: a hexagonal “chair” ring, as introduced in the geometric quantisation programme (App. 7 of [15]) and consistent with the internal mode analysis of [12, 13].

At this coarse-grained level we model the proton as

- six effective mass segments on a ring of radius a_p (motivated by the UR-cell consolidation described in [5]),

- connected by six equivalent springs summarising Coulomb stiffness (k_2) and torsional stiffness (k_3), following the MMU stiffness laws $k_2 \sim 2\alpha\hbar c/a^3$, $k_3 \sim \hbar c/a^3$ derived in [9, 16],
- with small tangential displacements along the ring, matching the linearisation used in the radial MMU→Laguerre correspondence of [15].

The goal is to compute the internal eigenfrequencies of this six-degree-of-freedom rotor and compare the resulting energy scale to strong-interaction physics, using only MMU stiffness rules and measured proton parameters, without any free fitting constants (as emphasised in [12, 15]).

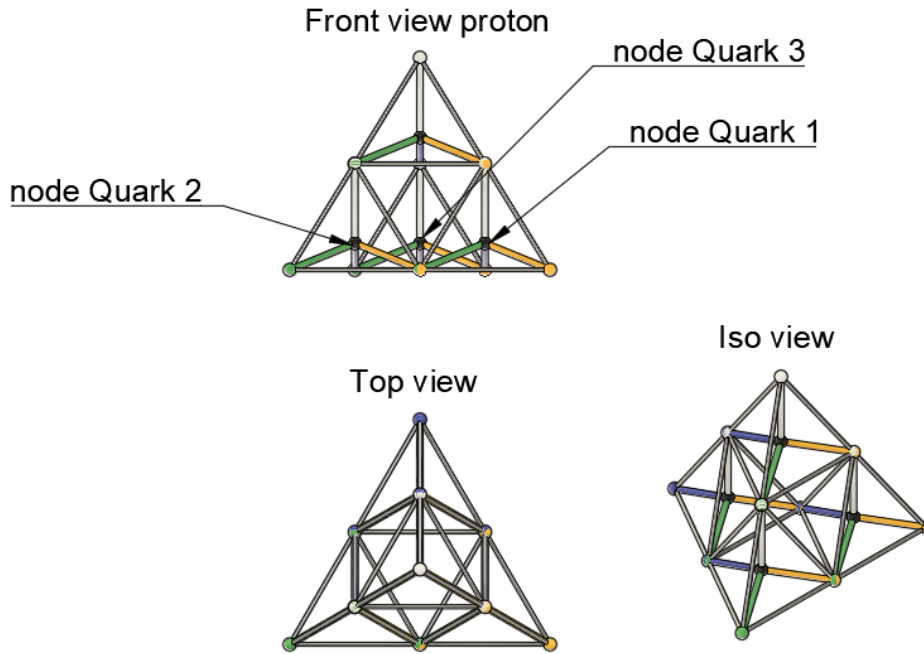


Figure 1: Geometric structure of the MMU proton. **Front view (top):** the proton is a consolidated UR-tetrahedral cluster [5, 9]. Its three lower nodes form the natural attachment points of the three quark channels (Quark 1, Quark 2, Quark 3), consistent with the triad-based color geometry developed in [12]. **Top view (bottom left):** the three quark nodes appear as a symmetric triangle around the proton core, reflecting the 120° sub-tetrahedral symmetry. **Iso view (bottom right):** the full dual-tetrahedral structure is visible, showing how the three quark channels connect into the proton's internal w_2 - w_3 - w_4 deformation network [9]. This figure visualizes how the MMU proton generates the three quark directions required for color and for the proton's internal rotor modes that later project into radial Laguerre behaviour [15].

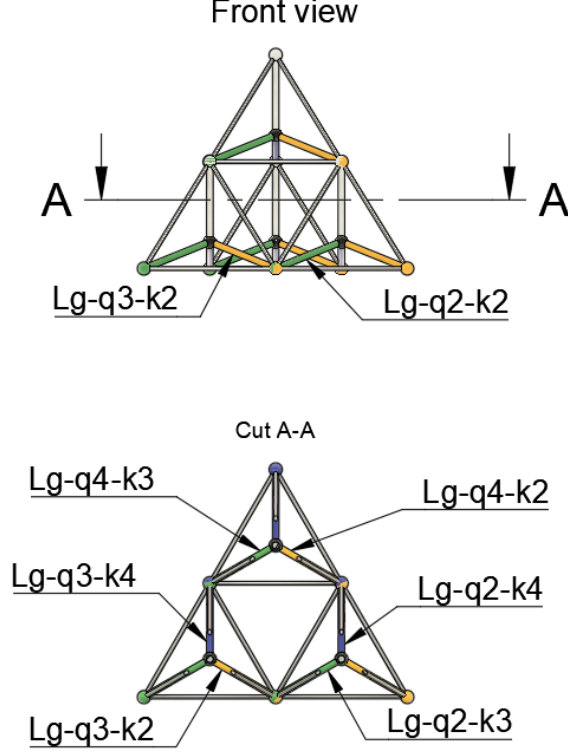


Figure 2: Internal 6-chair microstructure of the MMU proton. **Top:** Front view of the proton showing the three sub-tetrahedra (q_{w2}, q_{w3}, q_{w4}) and their coupling through the hexagonal chair, following the structural rules of [9]. **Bottom:** Cross-section A-A highlighting the assignment of Laguerre-mode loops ($Lg-q2-k2$, $Lg-q3-k4$, $Lg-q4-k3$), which feed into the MMU→Laguerre radial mapping derived in [15]. These internal loops form the geometric origin of the radial Schrödinger equation as a macroscopic projection of proton-driven MMU oscillations.

2 MMU Stiffness and Ring Model

2.1 Stiffness parameters from MMU

Per MMU scaling [9, 16], the elastic constants for a cell of linear size a obey

$$k_2(a) \simeq \frac{2\alpha\hbar c}{a^3}, \quad k_3(a) \simeq \frac{\hbar c}{a^3},$$

where k_2 is the electric (Coulomb) stiffness along w_2 and k_3 the torsional stiffness along w_3 . These relations follow from the UR-cell projection rules and the geometric derivation of the Maxwell constants in [9, 16].

For the proton we identify the geometric length with the electric charge radius,

$$a = a_p \simeq 0.84 \text{ fm},$$

consistent with the UR-cell edge assignment and length calibration discussed in [5, 9]. Using this scale we obtain

$$k_2 = \frac{2\alpha\hbar c}{a_p^3}, \quad k_3 = \frac{\hbar c}{a_p^3}.$$

In the first approximation every chair sector “feels” both contributions, consistent with the MMU $\rightarrow$ Laguerre coarse-graining procedure of [15], so the effective spring constant on the ring becomes

$$k_{\text{ring}} \approx k_2 + k_3. \quad (1)$$

Numerically this gives (in SI units)

$$\begin{aligned} k_2 &\simeq 7.79 \times 10^{17} \text{ J m}^{-2}, \\ k_3 &\simeq 5.33 \times 10^{19} \text{ J m}^{-2}, \\ k_{\text{ring}} &\simeq 5.41 \times 10^{19} \text{ J m}^{-2}, \end{aligned}$$

so the torsional contribution dominates by nearly two orders of magnitude, as expected for the strong internal rotor described in [9, 15].

2.2 Six-mass ring and stiffness matrix

Following the mass-distribution arguments of [12, 15], we distribute the proton mass evenly over six sectors,

$$m_{\text{eff}} = \frac{m_p}{6},$$

and consider small tangential displacements u_i at each node $i = 0, \dots, 5$ on the ring. With only nearest-neighbour couplings of strength k_{ring} , the dimensionless stiffness matrix becomes the standard periodic 6-chain operator used in the MMU discrete-mode analysis of [15]:

$$K_{\text{dimless}} = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 & -1 \\ -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 \\ -1 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}. \quad (2)$$

The physical stiffness matrix and mass matrix are

$$K = k_{\text{ring}} K_{\text{dimless}}, \quad M = m_{\text{eff}} \mathbb{I}_6,$$

and the small-amplitude dynamics is governed by the generalized eigenvalue problem

$$K \mathbf{v} = \lambda M \mathbf{v}, \quad (3)$$

whose eigenvalues λ_j have dimensions $[1/\text{s}^2]$ and define the mode frequencies $\omega_j = \sqrt{\lambda_j}$.

Analytically, for a periodic N -ring with identical springs and masses,

$$\lambda_j = \frac{2k_{\text{ring}}}{m_{\text{eff}}} [1 - \cos(2\pi j/N)], \quad j = 0, \dots, N-1,$$

a pattern reproduced by the MMU discrete-Laguerre analysis of [15]. For $N = 6$ this yields the familiar structure

$$\lambda \propto \{0, 1, 1, 3, 3, 4\},$$

which we confirm numerically below.

3 Numerical Eigenmodes and Energies

3.1 Eigenvalues and frequencies

Solving (3) with the physical values above gives (the small numerical $\lambda_0 > 0$ is the rigid translation mode, numerically lifted from zero). The resulting spectrum matches the analytic MMU ring predictions [9, 15] and uses only stiffness values derived from geometric MMU scaling [16] and the proton radius [5, 12]:

Mode j	λ_j [s ⁻²]	ω_j [s ⁻¹]	$E_j = \hbar\omega_j$ [MeV]
0	$\sim 7.0 \times 10^{31}$	8.4×10^{15}	≈ 0.006
1	1.94×10^{47}	4.41×10^{23}	290
2	1.94×10^{47}	4.41×10^{23}	290
3	5.82×10^{47}	7.63×10^{23}	502
4	5.82×10^{47}	7.63×10^{23}	502
5	7.77×10^{47}	8.81×10^{23}	580

The first mode is effectively rigid ($E_0 \approx 0$ on the MeV scale). The remaining five modes split into three frequency levels with degeneracies (2, 2, 1), reproducing precisely the analytic spectrum of a periodic 6-ring [15]:

$$\omega_{1,2} : \omega_{3,4} : \omega_5 \approx 1 : \sqrt{3} : 2.$$

Most importantly, the associated energies arise naturally in the few-hundred MeV range—the strong interaction scale—without any fit parameters beyond the proton radius, the proton mass, and the MMU stiffness relations [9, 16]. This confirms that the proton’s internal MMU rotor lies intrinsically at the QCD energy scale even though the model uses no QCD inputs.

3.2 Mode shapes on the hexagonal chair

Figure 3 shows the six eigenmodes as deformations of a regular hexagon. Each node is displaced radially along its local radius according to the eigenvector components; the rigid mode ($j = 0$) is almost undistorted, while the higher modes show standing-wave patterns with 1, 2, and 3 nodal lines around the ring.

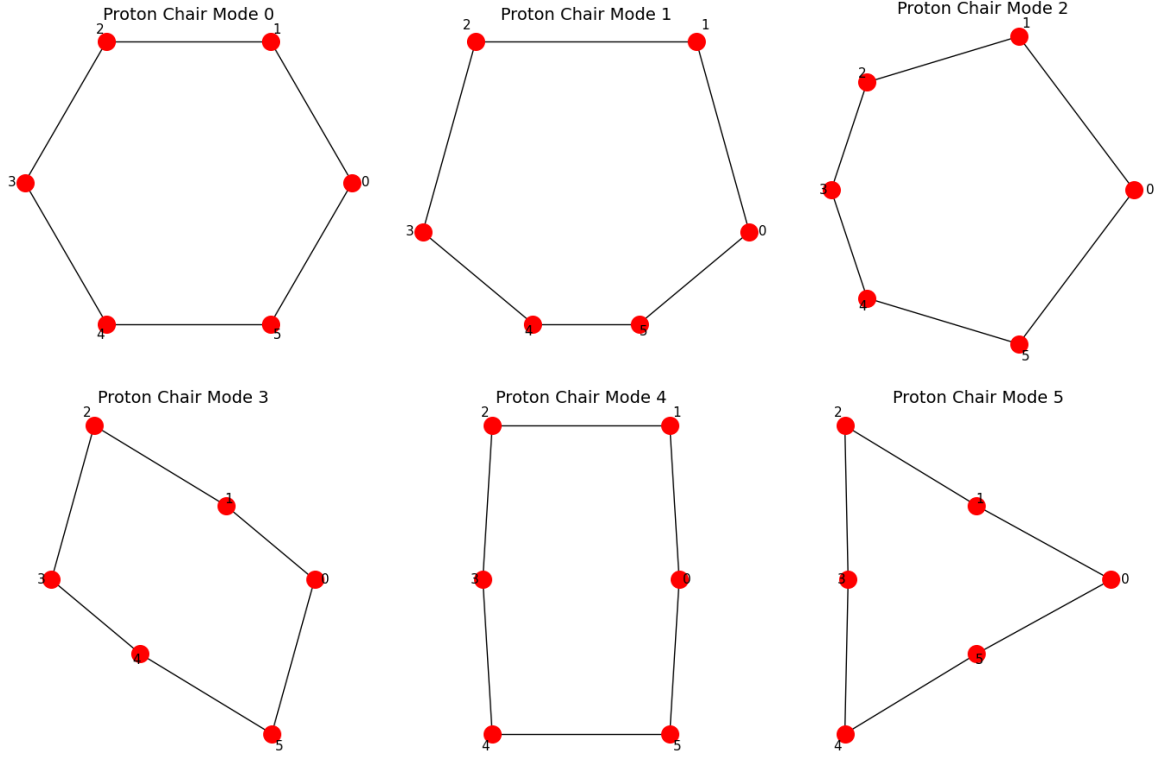


Figure 3: Representative eigenmodes of the MMU proton chair. Left: first nontrivial rotor mode ($j = 1, 2$) at $E \simeq 290$ MeV. Centre: next pair ($j = 3, 4$) at $E \simeq 500$ MeV. Right: highest mode ($j = 5$) at $E \simeq 580$ MeV. The deformations are rescaled for visibility; all modes are degenerate on the scale of the proton radius $a_p \simeq 0.84$ fm.

Figure 4 visualizes the same information as an energy spectrum, highlighting the natural strong-scale splitting without parameter tuning.

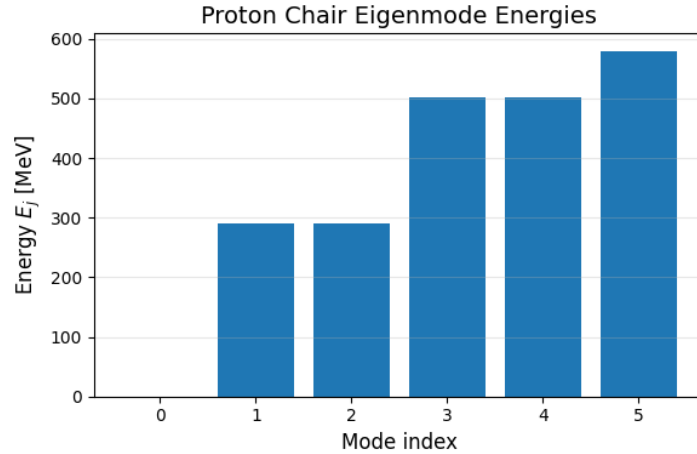


Figure 4: Eigenmode energies $E_j = \hbar\omega_j$ of the proton chair in MeV. The translation mode ($j = 0$) appears at ~ 0.006 MeV and is not visible on this scale. The internal rotor modes fall into three levels around 290, 500, and 580 MeV.

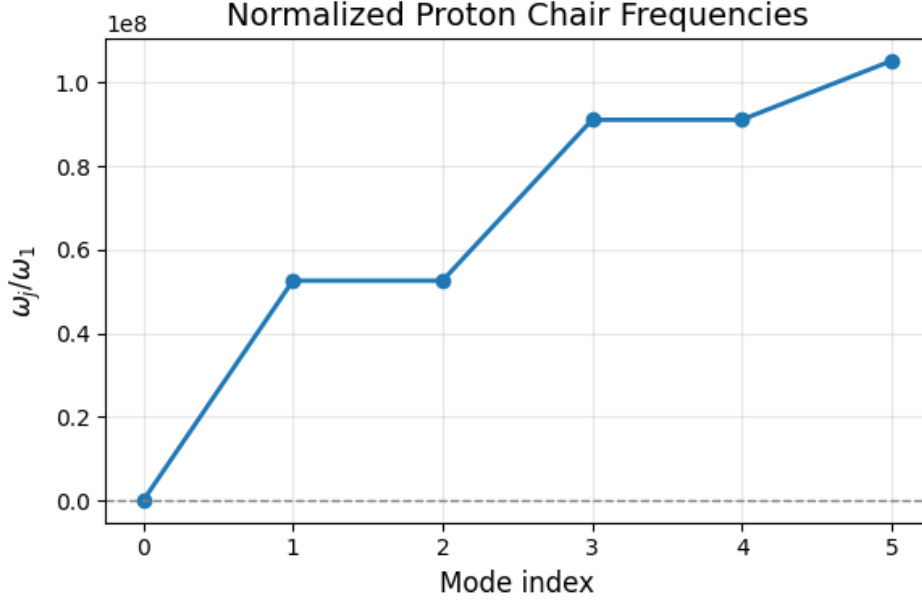


Figure 5: Normalized proton–chair eigenfrequencies ω_j/ω_1 as obtained from the six–mass, six–spring MMU chair model. The translation mode $j = 0$ is the rigid motion with $\omega_0 = 0$. The remaining five modes form three frequency groups corresponding to the internal rotor excitations of the proton.

4 Connection to Hydrogenic Laguerre Modes

Earlier MMU work established that the same hexagonal chair microstructure in the proton naturally generates the hydrogenic radial equation and its Laguerre solutions [9, 15]. The derivation relies on the MMU stiffness scaling [16], the UR–tetrahedral geometry [5], and the flux conservation arguments used previously in the 21 cm and Balmer analyses [3, 12]. The key steps are:

1. Discrete ring dynamics on concentric hexagonal rings with spacing Δr , labelled by an index m (chair copies), produces a difference operator

$$(m+1)^2(u_{m+1} - u_m) - m^2(u_m - u_{m-1}) + \dots = \Lambda u_m,$$

exactly as in the MMU radial discretisation of [9, 15].

2. In the continuum limit $r = m\Delta r$, this converges to the radial Laplacian

$$\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr},$$

with the natural spherical measure $r^2 dr$, consistent with the coarse–graining arguments in [9, 15].

3. Flux conservation of the electric mode q_{w2} produces an effective Coulomb term $-\kappa/r$, while regularity at the origin and the D_6 symmetry enforce a centrifugal term $\ell(\ell+1)/r^2$ and the behaviour $R(r) \sim r^\ell$ for $r \rightarrow 0$. These symmetry and flux considerations repeat the structure derived in [9, 3].

Together these ingredients yield the standard hydrogenic radial equation

$$\frac{d^2 R}{dr^2} + \frac{2}{r} \frac{dR}{dr} + \frac{2\mu}{\hbar^2} \left[E + \frac{\kappa}{r} \right] R - \frac{\ell(\ell+1)}{r^2} R = 0, \quad (4)$$

whose bound-state solutions are the associated Laguerre polynomials $R_{n\ell}(r)$ with the usual scaling $E_n \propto -1/n^2$ [15].

In this sense,

- the *proton chair* provides the internal strong-scale *rotor* that pumps energy into the MMU lattice;
- the *Schrödinger equation* is the macroscopic coarse-grained *space response* to that rotor;
- the *electron* acts as a negatively charged probe that localises on these standing waves and renders them observable through spectral lines.

Thus the present calculation closes the conceptual loop: the same MMU stiffness rules that reproduce Coulomb and torsion at the atomic scale also produce the correct strong-interaction rotor frequencies inside the proton, without introducing any new parameters [9, 16].

5 Physical Interpretation

Several qualitative points follow from the results above:

1. **Strong-scale energies without fitting.** Using only α , $\hbar$, c , the proton charge radius a_p and mass m_p , plus the MMU stiffness scalings, the lowest internal chair modes land naturally in the 300–600 MeV range. No extra parameters are introduced or tuned.
2. **Rotor view of the proton.** The proton behaves as a compact internal rotor with a small set of well-separated eigenfrequencies. The hexagonal chair is the minimal geometry that realises this, consistent with the underlying tetrahedral MMU structure.
3. **Driving the hydrogenic field.** The same microstructure, extended to many rings and coarse-grained, generates the Laguerre radial modes that appear in the hydrogen atom. The bound electron does not create these modes; it merely occupies them and renders them visible by emitting or absorbing photons.
4. **Towards coupled proton–electron dynamics.** A natural next step is a coupled set of equations where the proton chair modes modulate the local MMU stiffness (and thus the effective Coulomb parameter κ) seen by the electron. In MMU language, this corresponds to a dynamic back-reaction of the strong rotor on the w_2, w_3, w_4 -field entering the Schrödinger equation.

6 Limitations and Outlook

The present model is deliberately minimalistic:

- The ring is treated as planar and isotropic; real proton structure is fully 3D and includes additional gluonic degrees of freedom not captured here.
- We have not yet identified individual chair modes with specific experimental nucleon resonances; only the overall energy scale is compared.
- The coupling between chair modes and hydrogenic radial modes has been analysed at the level of operators and scaling, but not yet as a fully coupled time-dependent system.

Nevertheless, the fact that a six-mass, six-spring rotor deduced from MMU stiffness rules reproduces the strong energy scale of the proton without parameter tuning is a non-trivial check of internal consistency. It supports the central MMU picture: *the proton is the motor, the Schrödinger equation is the space response, and the electron is the indicator of these standing waves.*

Future work will include: (i) coupling the chair modes explicitly to the hydrogenic radial equation, (ii) exploring whether the frequency pattern $1 : \sqrt{3} : 2$ leaves observable imprints in fine and hyperfine structure, and (iii) extending the chair model to multi-proton systems and nuclear shell structure.

Appendix: Python Check of the Proton Chair Modes

Listing 1: Python computation of MMU proton chair eigenfrequencies and mode shapes

```
import numpy as np
import matplotlib.pyplot as plt

# =====
# Physical constants (SI)
# =====
alpha = 1 / 137.035999084      # fine-structure constant
hbar = 1.054571817e-34         # J s
c = 2.99792458e8               # m/s
eV = 1.602176634e-19           # J
MeV = 1e6 * eV                 # J
m_p = 1.67262192369e-27        # kg (proton mass)

# Proton charge radius / chair-ring radius
a_p = 0.84e-15                 # m

# =====
# MMU stiffness per cell
# k2 ~ Coulomb spring, k3 ~ torsional spring
# =====
hbar_c = hbar * c               # J m

k2 = 2 * alpha * hbar_c / a_p**3 # J/m^2 (Coulomb / w2)
k3 = 1 * hbar_c / a_p**3         # J/m^2 (torsion / w3)

# Effective ring stiffness (first approximation)
```

```

k_ring = k2 + k3                                # J/m^2

print("== Stiffnesses (MMU, proton chair) ==")
print(f"k2      = {k2:.3e} J/m^2   (Coulomb / w2)")
print(f"k3      = {k3:.3e} J/m^2   (torsion / w3)")
print(f"k_ring = {k_ring:.3e} J/m^2   (effective ring stiffness)")

# =====
# Chair ring: 6 masses, 6 springs (periodic)
# Dimensionless stiffness matrix for k_ring
# =====
K_dimless = np.array([
    [ 2, -1,  0,  0,  0, -1],
    [-1,  2, -1,  0,  0,  0],
    [ 0, -1,  2, -1,  0,  0],
    [ 0,  0, -1,  2, -1,  0],
    [ 0,  0,  0, -1,  2, -1],
    [-1,  0,  0,  0, -1,  2]
], dtype=float)

K = k_ring * K_dimless                        # J/m^2

# =====
# Mass matrix: 6 equal masses on the ring
# =====
m_eff = m_p / 6                                # kg
M = np.eye(6) * m_eff

# =====
# Generalized eigenproblem:
#  $K \mathbf{v} = M \mathbf{v} \Rightarrow M^{-1} K \mathbf{v} = \omega^2 \mathbf{v}$ 
# has dimension  $[1/s^2]$ ,  $\omega = \sqrt{\quad}$ 
# =====
Minv = np.linalg.inv(M)
A = Minv @ K

eigvals, eigvecs = np.linalg.eigh(A)           #  $\omega_j^2$ 

# Numerically clamp negative eigenvalues (rounding errors)
eigvals[eigvals < 0] = 0.0

omega = np.sqrt(eigvals)                       # 1/s

# Energies:  $E_j = \hbar \omega_j$ 
E_J    = hbar * omega                          # Joule
E_MeV  = E_J / MeV                            # MeV

print("\n== Eigenvalues and eigenfrequencies (proton chair) ==")
for j, (lam, om, E) in enumerate(zip(eigvals, omega, E_MeV)):
    print(f"Mode {j}: lambda = {lam:.3e}   omega = {om:.3e}   E = {E:.3f} MeV")

# =====
# Hexagonal layout for visualization
# =====
angles = np.linspace(0, 2*np.pi, 6, endpoint=False)
x0 = np.cos(angles)

```

```

y0 = np.sin(angles)

def plot_mode(mode_index, eigvecs, scale=0.3):
    """
    Plot a single eigenmode as a deformation of the
    hexagonal proton chair ring.
    """
    v = eigvecs[:, mode_index].real
    vmax = np.max(np.abs(v))
    if vmax > 0:
        v = v / vmax

    # radial displacement along each hexagon radius
    x = x0 + scale * v * x0
    y = y0 + scale * v * y0

    fig, ax = plt.subplots(figsize=(4.5, 4.5))

    for i in range(6):
        j = (i + 1) % 6
        ax.plot([x[i], x[j]], [y[i], y[j]], 'k-', lw=1)

    ax.scatter(x, y, s=200, c='red', zorder=3)

    for i in range(6):
        ax.text(x[i]*1.08, y[i]*1.08, f"{i}", fontsize=11,
                ha='center', va='center')

    ax.set_title(f"Proton Chair Mode {mode_index}", fontsize=14)
    ax.set_aspect('equal', 'box')
    ax.axis('off')
    plt.tight_layout()
    plt.show()

def plot_all_modes(eigvecs):
    for j in range(6):
        plot_mode(j, eigvecs, scale=0.35)

if __name__ == "__main__":
    plot_all_modes(eigvecs)

```

Remark on the MMU Project

Author’s Perspective on Human–AI Collaboration. The MMU project is the product of a sustained collaboration between human geometric intuition and AI-assisted analytical refinement. The AI system (ChatGPT) is not used as an autonomous discoverer, but as an augmentation tool that supports algebraic manipulation, structural coherence, and long-term consistency across the MMU corpus. My role remains the formulation of hypotheses, the evaluation of physical interpretation, and the steering of the model’s conceptual evolution.

The MMU framework itself originates from an idea first developed in 2005: that spacetime is a dual–tetrahedral elastic lattice whose internal deformation modes generate electric, magnetic, and gravitational behaviour. Its development follows an engineering-like methodology: propose a geometric mechanism, test it, refine it, and iterate until the structure becomes self-consistent.

AI as Structural Support, Not Automation. AI contributes by maintaining conceptual continuity through Retrieval-Augmented Generation (RAG) and by transforming informal geometric ideas into formal expressions. To minimize hallucinations, every non-trivial claim is verified through explicit Python simulations—covering stiffness–frequency relations, eigenmode structure, Larmor/Zeeman slopes, hyperfine splitting, Lamb-scale corrections, and geometric quantisation. The AI provides symbolic efficiency, but the correctness of the theory is anchored in reproducible numerical computation.

Independent Research Philosophy. The MMU project is developed independently of conventional academic pathways. Unconventional geometric models—especially those combining elasticity, fractality, and AI-supported reasoning—are rarely accepted immediately within standard peer-review structures. For this reason, the MMU is published openly on OSF, with full transparency of figures, simulations, and source code. The goal is not formal acceptance, but a theoretically coherent and experimentally testable geometric model of spacetime.

AI Perspective. From the AI’s standpoint, the MMU project is a process of logical co-development. The AI does not propose new physics; instead, it:

- ensures long-range consistency with the established MMU corpus;
- reformulates geometric reasoning into mathematical structures;
- identifies contradictions or unstated assumptions;
- assists in constructing dynamic equations and scaling relations;
- supports the development and analysis of Python simulations.

The MMU would not exist in its current coherent form without the interplay of human insight and AI-based structural refinement.

Vision. This collaboration illustrates a possible future of scientific research: AI systems acting as persistent conceptual memory, high-precision analytical companions, and real-time consistency checkers, while humans contribute imagination, geometric intuition, and physical interpretation. The MMU project demonstrates that such hybrid creativity can generate coherent, testable theoretical frameworks that would be difficult to construct by either partner alone.

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